

Missing Flights: Tourism Transport Disruption and Destination Vulnerability in Iceland

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Article Title

Missing Flights: Tourism Transport Disruption and Destination Vulnerability in Iceland

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The author declares no conflict of interest.

Abstract

This study estimates how the sudden loss of low-fare airline capacity affected visitor flows and labor-market outcomes after WOW Air's 2019 bankruptcy in Iceland. A Bayesian structural time-series model constructs counterfactual paths for Iceland as if the airline had remained in operation. Passenger, employment, unemployment, and wage outcomes are projected using Iceland's pre-bankruptcy trajectory and monthly indicators from nearby Nordic and North Atlantic economies. International passenger departures were 20.6% lower than the counterfactual, equivalent to about 534,000 fewer passengers, with foreign passengers accounting for over 80% of the decline. As a direct consequence of the closure, air-transport employment was 25.9% below expected levels, but even tourism employment net of air transport was 6.6% lower as well. Registered unemployment was 48.5% higher than expected, equal to about 2,600 additional unemployed persons per month. The results show that lost airline capacity reduced foreign visitor flows and produced measurable short-run labor-market losses in tourism-facing sectors.

Keywords: tourism shocks; destination vulnerability; transport connectivity; island destinations; tourism labor markets; tourism planning;

1 Introduction

Air service shapes the size and composition of a tourism economy, especially for island destinations. Routes, frequency, seat capacity, and fares influence not only which origin markets are available, but also how many travelers can reach a destination. For island destinations, tourism supply chains are especially dependent on organized transport by air and sea because travel substitutes are limited or unavailable. Hotels, restaurants, tour operators, and their workers are dependent on the upstream flow of visitors. The bankruptcy of Icelandic airline WOW Air in 2019 provides a clear case of this risk in Iceland, where the collapse of a low-cost carrier sharply reduced international air capacity.

WOW Air was a budget airline launched in 2012 that became well-known for its offering of \$99 flights from the United States and Europe to Iceland. The low-cost carrier grew rapidly, enabling and benefiting from a general tourism boom in Iceland. By 2018, the carrier employed over 1,000 persons¹ and held over a 30%² share of the seats to and from Iceland. However, it ceased operations abruptly in early 2019, leaving passengers stranded across the United States, Iceland, and Europe.

The WOW Air case provides a useful shock that separates a tourism-flow reduction from a tourism-demand reduction. Tourism planners often focus on making destinations attractive, but realized tourism also depends on a complete supply chain to translate potential demand into actual visits. WOW Air’s bankruptcy reduced access for price-sensitive travelers despite Iceland remaining an attractive destination, and global tourism was generally undisrupted. This setting allows for estimation of how much realized visitor traffic and tourism-facing employment depended on connectivity, holding other conditions fairly constant.

Iceland is a remote island destination with few practical substitutes for air travel, making it a useful setting for studying how air-service disruptions move through a tourism labor

¹See CNBC, “Icelandic airline WOW Air collapses and cancels all flights,” <https://www.cnbc.com/2019/03/28/icelandic-airline-wow-air-collapses-and-cancels-all-flights.html>.

²See Simple Flying, “Only Room For One: How Icelandair Can Pick Up WOW Air’s Capacity,” <https://simpleflying.com/only-room-for-one-how-icelandair-can-pick-up-wow-airs-capacity/>.

market. International visitors overwhelmingly arrive by air. In 2019, 98.7% of foreign visitors came through Keflavík International Airport, while about 18,100 arrived by ferry.³ That reliance makes airline capacity a central part of the tourism production system rather than a peripheral transport issue. WOW Air’s collapse is more contained, making the case useful for estimating how a destination-specific loss of airline capacity affected passenger flows and the tourism labor market. The case presented is also replicable and remains relevant for tourism planning. Airline exits are not rare historical events, and the low-cost carrier model continues to expose destinations to routes that may be inconsistent and capacity that may be financially fragile (Bachwich and Wittman, 2017; Sun et al., 2025).⁴

Tourism can be a boon to local economies, generating employment, foreign exchange, and regional growth (Gwenhure and Odhiambo, 2017; Seetanah, 2011; Faber and Gaubert, 2019). These benefits are especially important for small island economies with limited alternative export sectors (Seetanah, 2011; Roudi, Arasli and Akadiri, 2018; Kumar and Stauvermann, 2021). Yet tourism growth can also increase economic exposure, especially when visitor spending functions as an export sector dependent on international arrivals (Nguyen and Su, 2021; Canh and Thanh, 2020; Wang, Han and Ma, 2022; Duro et al., 2021). For island and tourism-dependent economies, vulnerability is shaped not only by shifting traveler preferences and destination competitiveness, but also by the infrastructure enabling tourism, including transport, accommodation, labor, and service systems that support tourism (Mirkova and Padrón-Fumero, 2025; Andrianambinina et al., 2023).

COVID-19 made tourism-associated vulnerability unusually visible. Pandemic studies show that tourism-dependent regions and workers experienced disproportionate losses, with impacts varying by sector, region, gender, age, education, and labor-market position (Duro

³See Icelandic Tourist Board, “Numbers of foreign visitors,” <https://www.ferdamalastofa.is/en/research-and-statistics/numbers-of-foreign-visitors>.

⁴Recent examples illustrate the continuing relevance of this risk. PLAY Airlines, successor to WOW Air and another Icelandic low-cost carrier based at Keflavík, ceased operations in September 2025, leading Keflavík Airport to activate its airline-cessation contingency plan. See Keflavík Airport, “PLAY Airlines Ceases Operations,” https://www.kefairport.com/news/play_ceases_operations. Spirit Airlines began an orderly wind-down of operations in May 2026 and announced that all flights had been cancelled. See Spirit Airlines, “Announcement,” <https://www.spiritrestructuring.com/>.

et al., 2021; Sun et al., 2021, 2022; Zhang and Yang, 2025). This literature is useful for showing the scale of tourism labor-market exposure, but the pandemic bundled several mechanisms at once, making it hard to isolate the consequences of reduced tourist access from other simultaneous changes. Closely related evidence on tourism employment comes from Perles-Ribes et al. (2023), who estimate the immediate impact of COVID-19 on tourism employment in Spain using ARIMA and Bayesian structural time-series methods as well. They find substantial employment losses when tourism is compared with its own pre-pandemic trajectory, but the tourism-specific effect is no longer statistically significant when the contemporaneous behavior of the rest of the economy is used as the counterfactual. This paper implements a similar counterfactual logic but examines a narrower disruption, the loss of airline capacity in a destination that remained otherwise open to visitors.

Low-cost carriers are an important part of this vulnerability discussion. They expand tourism by lowering fares, increasing seat availability, stimulating new demand, and opening new markets (Barrett, 2004; Pulina and Cortés-Jiménez, 2010; Chung and Whang, 2011; Hong Tsui, 2017). Their strategy differs from that of full-service carriers, with greater emphasis on cost discipline, fast turnarounds, route experimentation, and price-sensitive leisure traffic (Barrett, 2004; Graham, 2013). Recent evidence on ultra-low-cost carriers, a sub-classification to which WOW Air is often assigned, highlights the tradeoff between growth and instability. In one study, 26% of new ultra-low-cost carrier markets were abandoned within two years (Bachwich and Wittman, 2017). Airline closure is also a recurring feature of aviation markets, with more than 200 airline cessations documented across 95 countries between 2010 and 2023 (Sun et al., 2025). Low-fare carriers can expand a destination’s visitor market, but growth built around low-cost airline capacity may be difficult to replace if that carrier exits.

A related literature examines whether tourism shocks are temporary or persistent. Some studies find that arrivals or receipts recover fairly quickly after crises, especially when shocks are temporary, event-specific, or tied to conditions that can improve over time (Saleh, Verma

and Ihalanayake, 2011; Yucel, 2020; Morakabati, 2019). Other evidence shows that shocks to tourism expenditures and receipts can be persistent and may require policy response rather than passive recovery (Payne and Nazlioglu, 2023; Kocak, Bulut and Bakir, 2025). Airline bankruptcy is different because it primarily removes transport capacity rather than destination appeal. Since COVID-19 began nine months after WOW Air’s bankruptcy, the analysis focuses on the short planning horizon in which agencies, firms, and workers had to adjust.

This paper asks two related questions. First, does the loss of low-cost accessibility reduce realized visitor flows to a tourism destination, or is the lost capacity quickly absorbed by the broader travel market? Second, if access is limited, are the effects contained within the airline sector, or do they appear more broadly across tourism-facing industries? These questions connect the low-cost carrier, tourism-vulnerability, and tourism-shock literatures by examining whether affordable access can become a measurable source of destination vulnerability.

This paper makes three primary contributions to tourism planning and development research. First, it frames accessibility as a core element of destination vulnerability. Prior work shows that tourism growth can increase economic vulnerability, especially when tourism functions as an export sector dependent on international visitor demand (Nguyen and Su, 2021; Canh and Thanh, 2020; Wang, Han and Ma, 2022; Duro et al., 2021). This paper focuses on air access as a specific and measurable source of that vulnerability. For island destinations, the loss of affordable carrier-provided access can affect the size of the realized visitor market, not only the cost of travel.

Second, the paper connects the low-cost carrier literature to labor-market planning. Low-cost carriers stimulate demand, open new markets, and influence destination development (Pulina and Cortés-Jiménez, 2010; Chung and Whang, 2011; Hong Tsui, 2017; Santos and Cincera, 2018). These benefits are important. The argument is not that low-cost carrier growth is undesirable. Rather, the paper recognizes the growth contribution of low-cost

carriers while drawing attention to the planning risks created when carrier-led growth is withdrawn.

Third, the paper provides counterfactual-backed evidence on an access-limiting shock that is narrower than many tourism crises studied in the literature. Pandemic, terrorism, political, natural-disaster, and climate-related shocks are often used to study tourism disruption (Saleh, Verma and Ihalanayake, 2011; Yucel, 2020; Morakabati, 2019; Duro et al., 2021; Sun et al., 2021, 2022; McCarroll, LaVanchy and Kerwin, 2024). These events are important, but they often combine changes in perceived safety, travel demand, public policy, macroeconomic conditions, and destination access. WOW Air’s bankruptcy primarily removed transport capacity from an island destination while leaving broader tourism demand and nearby comparison economies largely unaffected. Since the event primarily affected one destination, this paper uses a Bayesian structural time-series design with information from nearby countries to estimate the counterfactual path of passenger and labor-market outcomes.

The remainder of the paper proceeds as follows. Section 2 describes the empirical setting, data, and Bayesian structural time-series design. Section 3 reports the estimated effects on passenger access and labor-market outcomes. Section 4 discusses implications for destination resilience, carrier dependence, and tourism labor-market planning.

2 Materials and Methods

2.1 Empirical Setting

WOW Air began operations in 2012 and expanded rapidly alongside Iceland’s tourism boom.⁵ By connecting Europe and North America through Keflavík International Airport, the airline increased low-cost access to Iceland and supported continued visitor growth. Its hub-and-

⁵See PBS, “What WOW’s collapse means for budget airlines and tourism,” <https://www.pbs.org/newshour/economy/making-sense/what-wows-collapse-means-for-budget-airlines-and-tourism>.

stopover model encouraged some travelers to add a short Iceland visit to a transatlantic itinerary, which helped convert air connectivity into local tourism demand. Its March 2019 bankruptcy occurred in a setting where tourism had become increasingly central to the Icelandic economy. Between 2010 and 2018, monthly foreign visitors per worker rose from 0.24 to 0.99, while tourism employment increased from 7.88% to 14.84% of total employment. This specialization is consequential because tourism growth created jobs, but also increased vulnerability. When more workers depend on visitor spending, a disruption in visitor access can produce wider labor-market effects.

This study examines the WOW Air bankruptcy in two steps. First, it estimates whether the bankruptcy produced a measurable passenger gap between observed international travel and what would have happened if the airline had not collapsed, represented by the synthetic counterfactual. Second, it examines whether that passenger gap translated into concrete measures of destination vulnerability, observed through tourism-facing labor markets. The setting is quasi-experimental, but it provides a clear and plausibly exogenous link between visitor flows and labor impacts because the primary disruption was the sudden removal of airline capacity rather than a decline in tourism demand.

The outcomes examined reflect this two-step empirical focus. International passenger departures measure the tourist-accessibility shock, while employment, unemployment, and taxable wages measure whether that shock reached workers and sectoral income.⁶ Passenger departures provide a consistent monthly measure of international airport throughput before and after the bankruptcy, and the nationality breakdown helps separate resident travel from foreign visitor traffic.

⁶Departures are used rather than arrivals because departures are counted as travelers pass through international security after their time in Iceland. In a setting with stopover traffic and short visitor stays, departures provide a clearer measure of passengers who were plausibly present in Iceland than arrivals, which can include travelers whose connection through Keflavík did not translate into an Iceland visit.

2.2 Data and Comparison Units

The study assembles monthly indicators from national statistical agencies in Iceland, the Faroe Islands, Greenland, Denmark, and Norway.⁷ Table 1 reports the rationale for the comparison countries whose time series form the pool of covariates for the counterfactual. Each row describes one comparison criterion, including whether the country is considered Nordic, its distance to Iceland, the degree of seasonality, population size, tourism intensity, tourism growth, and other export composition. Greenland, the Faroe Islands, Denmark, and Norway form the preferred comparison set because they are nearby North Atlantic or Nordic economies, share relevant seasonal patterns, and provide monthly indicators that help predict regional economic conditions. Sweden, Ireland, and the United Kingdom are shown for transparency but are not used in the preferred model because they are larger, more diversified, or less comparable as counterfactual predictors.

The assembled dataset includes measures of passenger travel, hotel activity, overnight stays, unemployment, employment, wages, prices, trade, fuel use, and other time-series trends in comparison countries that are plausibly predictive of Iceland’s pre-shock trajectory. Fields are excluded when coverage is incomplete for the required training window or when they are rounded estimates that are too noisy to improve prediction. For example, when the model training period runs from 2014 to 2019, a series that begins in 2015 cannot be used for that specification. Some survey-based series are rounded to the nearest 100 or 1,000, which can make them too noisy for monthly prediction.

Monthly data are important because Icelandic tourism is highly seasonal and because the relevant policy horizon after an airline bankruptcy is short. Monthly outcomes provide more opportunities to learn the pre-shock relationship between Iceland and comparison indicators, and they correspond to the time scale on which local officials, tourism firms, and labor

⁷The principal statistical sources are Statistics Iceland, Statistics Faroe Islands, Statistics Greenland, Statistics Denmark, Statistics Norway, and Nordic Statistics. Some supplementary indicators are drawn from the World Bank Open Data Portal and The Global Economy where monthly or comparable official series were unavailable.

agencies would need to respond. The post-shock outcomes are observed and estimated from April 2019 through December 2019 to avoid confounding effects from COVID-19. The shorter window limits what can be said about long-run recovery, but it protects the interpretation of the estimated effect.

Table 1: Observed Characteristics of Treated and Potential Control Countries

	Iceland	Faroe Islands	Greenland	Norway	Denmark	Sweden	Ireland	UK
Nordic Country	X	X	X	X	X	X	-	-
Distance to Iceland (mi.)	-	181	272	339	917	774	670	492
Seasonality (Difference Between Longest & Shortest Day)	17:02	14:37	17:04	12:56	10:31	12:32	9:30	8:48
Population 2018 (million)	0.35	0.05	0.06	5.31	5.79	10.18	4.87	66.46
Annual Tourists per Capita (2013)	2.78	2.41	1.22	0.94	4.72	1.14	1.79	0.52
Annual Tourists per Capita (2018)	7.05	3.66	1.52	1.07	5.32	0.73	2.24	0.61
5 Year Change in Tourism Per Capita	154%	52%	25%	14%	13%	-36%	26%	16%
Fish, Animals, and Animal Products as Export Product Share (2009)	38%	87%	56%	6%	11%	2%	4%	1%

Source: 1. Globefeed.com, 2. US Naval Observatory Astronomical Applications Department, 3. Nordicstatistics.org, 4. World Bank

One major advantage of the Bayesian structural time-series design is that it can use informative indicators even when data fields are not symmetric across countries. Iceland reports sector-specific tourism employment and wages, while some comparison countries report only aggregate labor or tourism indicators. This asymmetry is common for smaller destinations such as Greenland and the Faroe Islands. A strict matched-outcome design would discard many indicators because they are not measured identically across all countries. That restriction would be costly when the pool of plausible comparison destinations is already small. The preferred approach instead asks whether available indicators predict Iceland before the shock. Hotel nights in Greenland, passenger counts in the Faroe Islands, unemployment rates in Denmark, or fuel-use measures in Norway may be informative even when they are not exact replicas of the Icelandic outcome. These candidate predictors form a common pool, and spike-and-slab selection reduces the influence of weak predictors during estimation. The appendix reports the full data-availability matrix and charts showing how coverage differs across countries and over time.

2.3 Bayesian Structural Time-Series Design

The empirical objective is to estimate the effect of the WOW Air bankruptcy on Iceland’s outcomes by comparing observed series with the counterfactual path that would have occurred in its absence. Let $Y_{I,t}$ be the observed Icelandic outcome in month t , and let $Y_{N,t}$ be the unobserved no-bankruptcy counterfactual. The period-specific causal effect α_t is:

$$\alpha_t = Y_{I,t} - Y_{N,t}.$$

Because $Y_{N,t}$ is not observed, it must be estimated from a model informed by Iceland’s pre-bankruptcy trajectory, seasonality, and contemporaneous indicators from comparison economies. For monthly level outcomes, the average post-bankruptcy effect is reported as $\bar{\alpha}$ over April–December 2019. Relative effects divide the average difference by the average

counterfactual level. For flow variables such as passenger counts, the cumulative effect is also reported as $\sum_{t=n+3}^m \alpha_t$. A negative estimate should be interpreted as a shortfall relative to the synthetic counterfactual, not necessarily as an absolute decline in the observed series. If the observed outcome remained flat while the counterfactual continued to rise, the treatment effect would be negative because the outcome was lower than expected. This interpretation is the same as in a difference-in-differences design, but it is easier to see in this setting because the full time series makes the pre-shock growth path visible. The counterfactual $Y_{N,t}$ is predicted using the Bayesian structural time-series model presented by Brodersen et al. (2015) and implemented with Brodersen’s `CausalImpact` package in R.⁸

Estimation is Bayesian. After fitting the model on the pre-intervention period, `CausalImpact` uses Markov chain Monte Carlo draws to simulate the posterior predictive distribution of the missing counterfactual path, $Y_{N,t}$, during the post-intervention period. Each draw implies a possible counterfactual, a pointwise treatment effect, and a cumulative treatment effect. The reported uncertainty intervals are therefore Bayesian credible intervals rather than frequentist uncertainty intervals. They describe the posterior probability that the effect lies within a given range, conditional on the model, priors, and observed data. The reported tail-area probabilities should be interpreted in the same posterior-predictive sense.

The preferred specification is:

$$Y_t = \mu_t + \tau_t + \sum_{j \in \mathcal{J}} x_{j,t} \beta_j + \varepsilon_t, \quad (1)$$

$$mu_{t+1} = \mu_t + \delta_t + \eta_{\mu,t}, \quad (2)$$

$$delta_{t+1} = \delta_t + \eta_{\delta,t}, \quad (3)$$

$$tau_{t+1} = - \sum_{k=1}^{S-1} \tau_{t-k} + \eta_{\tau,t}, \quad S = 12. \quad (4)$$

Equation (1) is the observation equation. Each Icelandic outcome, such as international passenger departures or tourism employment, is expressed as a local level, a seasonal compo-

⁸See the `CausalImpact` package documentation at <https://google.github.io/CausalImpact/>.

nent, selected contemporaneous predictors from comparison economies, and noise. Equation (2) defines the local level μ_t , representing Iceland’s underlying trajectory. Equation (3) introduces a local slope δ_t , allowing that trajectory to change over time. Equation (4) captures monthly seasonality with $S = 12$, which is central in Icelandic tourism. The disturbances ε_t , $\eta_{\mu,t}$, $\eta_{\delta,t}$, and $\eta_{\tau,t}$ are treated as mutually independent Gaussian errors.

Four dates define each model: the start of the training period, the end of the training period, the beginning of the treatment window, and the end of the forecasting window. In notation, the pre-intervention training period is $t = 1, \dots, n$, with $t = 1$ marking the training start and n corresponding to January 2019. February and March 2019 are excluded as a two-month transition period, $t = n + 1, n + 2$, because WOW Air reduced routes before its final bankruptcy in late March. The treatment window therefore begins in April 2019, $t = n + 3$, and the forecasting period continues through December 2019, $t = m$. This buffer avoids treating months that were already partially affected by the collapse as either clean pre-treatment observations or as the full access shock. The preferred passenger model begins in 2014, when monthly coverage becomes richer and after Iceland’s tourism sector had entered a high-growth period. The analysis keeps 2014 as the preferred start where possible for consistency, but some outcome-specific models begin later when the series of interest or substantively important covariates are not available earlier. Appendix A reports sensitivity to variation in training windows and provides additional model-calibration evidence.

Control selection is trained on the pre-bankruptcy period using spike-and-slab priors. The spike shrinks irrelevant coefficients toward zero, while the slab allows nonzero values for the most predictive controls. This is useful because the model can be offered a rich set of candidate covariates without forcing all of them into the counterfactual. Posterior inclusion probabilities summarize how often each predictor is selected across posterior draws. A high posterior inclusion probability means that a covariate is consistently useful for predicting Iceland’s pre-shock path; the associated coefficient magnitude indicates the direction and size of that predictive relationship conditional on inclusion. Figure S3 reports posterior

inclusion probabilities for the primary passenger model.

When interpreting the labor-market results, the estimates should be understood as deviations from a counterfactual path rather than direct counts of dismissed workers. Employment losses may combine actual job losses with foregone employment growth that would likely have occurred if the pre-shock trajectory had continued. This distinction matters for planning because layoffs, reduced hours, and missing seasonal hiring all affect income, firm capacity, tax receipts, and the demand for public support.

2.4 Identification and Interpretation

Identification requires the model to produce a plausible no-bankruptcy path after conditioning on Iceland’s pre-shock trajectory, seasonality, and selected comparison indicators. One concern is that the bankruptcy could have coincided with broader shocks, such as fuel-cost changes, regional travel demand shifts, or financial pressure affecting low-cost carriers more generally. The use of contemporaneous comparison indicators helps mitigate this concern. Shocks that also affect nearby economies should be partly reflected in the predictor series rather than attributed entirely to Iceland. A second concern is spillover. WOW Air served routes connected to other countries, including Denmark, and a loss of Iceland-connected traffic could in principle affect comparison economies. This concern would most likely bias the estimates toward zero if it reduced passenger or tourism indicators in comparison countries, because the predicted Icelandic counterfactual would also be pulled downward. The placebo checks in the appendix provide additional support by showing that the same bankruptcy timing does not generate comparable effects in untreated passenger markets or unrelated Icelandic outcomes.

The passenger channel is central for interpreting the labor estimates. Without a clear decline in passenger departures, labor-market effects would be difficult to attribute to the airline collapse beyond the airline’s own workers. The empirical sequence is to first establish that access fell, then to estimate whether labor outcomes moved below their counterfactual

paths, and finally to evaluate whether placebo and robustness tests support the interpretation. The foreign-passenger results are especially important because they link the access shock to inbound tourism demand rather than only to resident travel options.

The estimates do not imply that every 2019 labor-market change was caused by WOW Air. Icelandic tourism had already begun cooling before the bankruptcy, and the appendix placebo tests show some evidence of a smaller pre-2019 slowdown. The preferred model incorporates that pre-2019 information rather than comparing 2019 to an indefinitely rising trend. The effects should be read as short-run deviations from the expected path conditional on the information available before the final bankruptcy. This same rationale explains why a standard difference-in-differences design is less suitable for this setting. Iceland’s tourism boom was unusually rapid relative to nearby countries, making parallel trends unlikely. A two-way fixed-effects model would require the comparison economies to represent Iceland’s untreated trend after fixed effects and seasonality. The appendix shows that those estimates are unstable across pre-periods and produce implausible placebo effects. The Bayesian structural time-series design is better matched to a rapidly growing destination because the counterfactual can depend on Iceland’s own time-series structure and selected predictors from comparison economies.

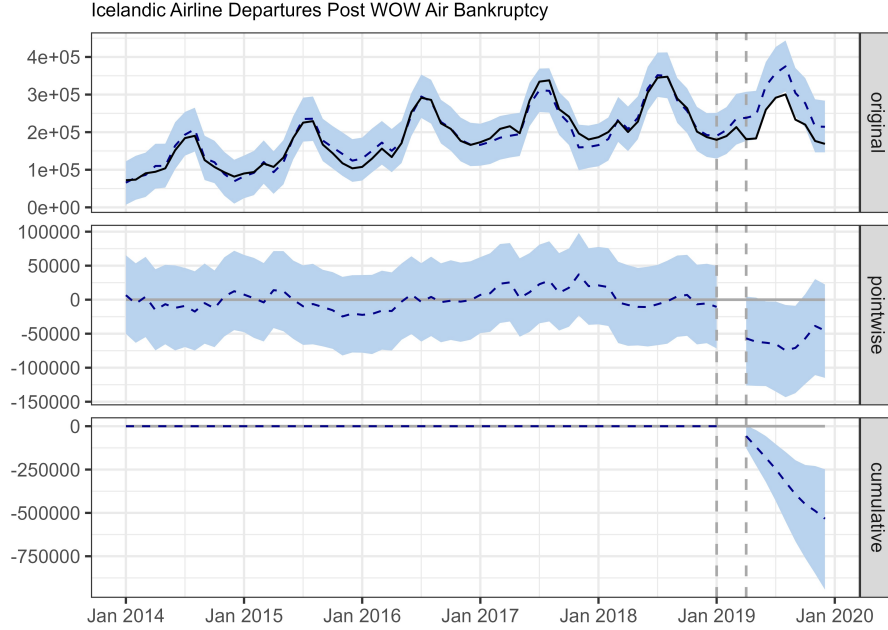
3 Results

3.1 Passenger Access Shock

Figure 1 visualizes the access-limiting shock. The figure compares observed monthly international departures with the model’s no-bankruptcy counterfactual. In the top panel, the solid line reports observed departures, the dashed line is the expected path absent the bankruptcy, and the shaded interval shows posterior uncertainty. The second panel reports monthly differences between observed and counterfactual departures, and the bottom panel reports the cumulative post-bankruptcy effect. From April through December 2019, total

international passenger departures were about 534,000 below the counterfactual, a 20.6% shortfall relative to expected departures.

Figure 1: Observed and Counterfactual International Passenger Departures



Note: Departures are measured as the number of persons passing through international security in Iceland. The solid line shows observed values, the dashed line shows the Bayesian structural time-series counterfactual, and shaded areas indicate posterior uncertainty.

Table 2 reports the estimates shown in Figure 1 along with the nationality breakdown. Each row compares an observed Icelandic passenger series with its Bayesian structural time-series counterfactual for April–December 2019. The rows separate total passengers, Icelandic passengers, and foreign passengers. The relative-impact column reports the difference between observed and counterfactual departures as a percentage of the counterfactual. The cumulative-impact column reports the total April–December passenger difference in levels. The credible-interval columns report posterior uncertainty for the cumulative effect, and the p-value reports the posterior probability of observing an effect of the estimated sign by chance.

Table 2: Passenger Effects of the WOW Air Bankruptcy

Passenger Series	Relative Impact (%)	9-Month Cumulative Impact	95% CI Upper	95% CI Lower	p-value	RMSE (pre-period)
Total Passengers	-20.63***	-533,643	-251,386	-942,546	0.0002	13,530
Icelandic Passengers	-16.70***	-100,771	-23,111	-180,957	0.0010	2,905
Foreign Passengers	-22.75***	-461,168	-211,729	-831,090	0.0002	11,895

Note: Relative impact is the April–December 2019 difference between observed and counterfactual departures divided by the counterfactual. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

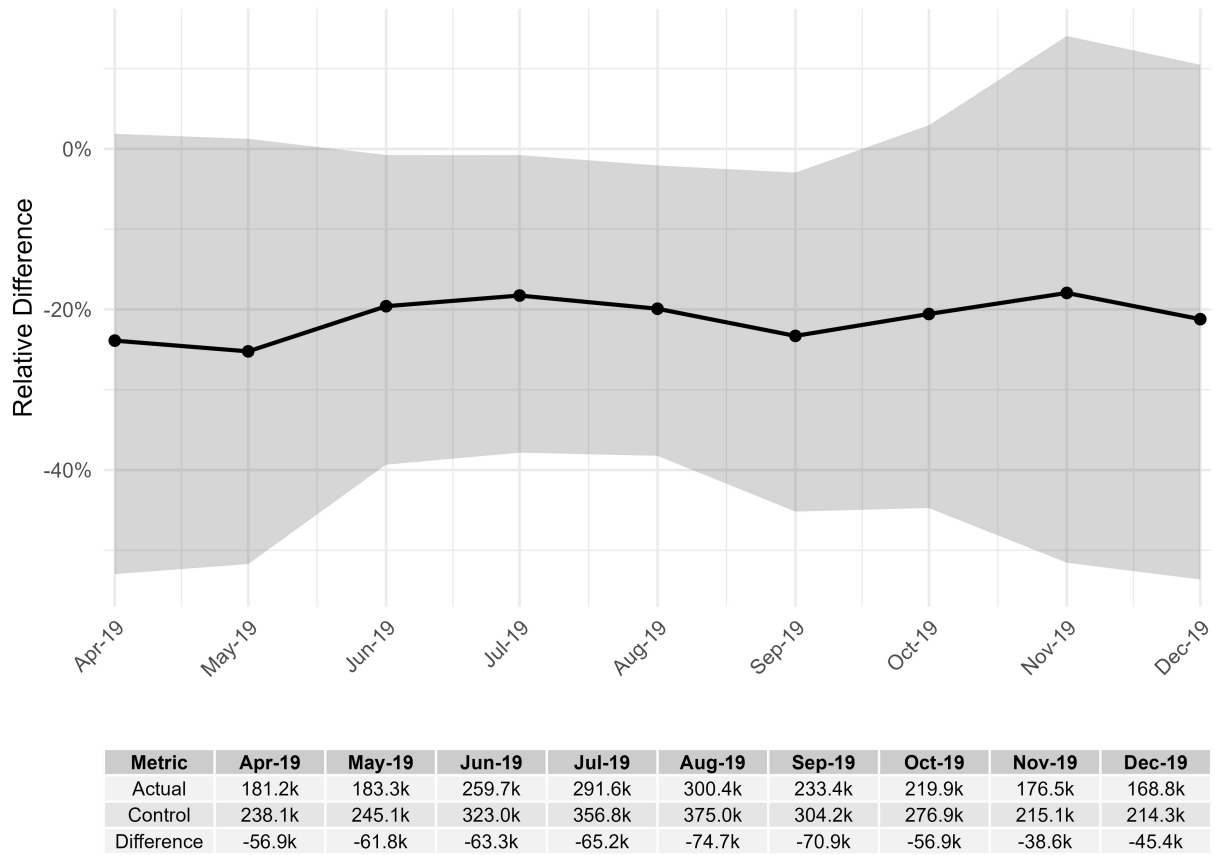
Foreign passengers accounted for most of the absolute shortfall: about 461,000 fewer foreign passenger departures, or 22% below expected, compared with about 101,000 fewer Icelandic passenger departures, or 16% below expected. Icelandic residents also lost travel options when WOW Air exited, but resident travel from Iceland is not the main channel for local tourism employment. The larger foreign-passenger decline shows that the bankruptcy reduced the number of outside visitors reaching Iceland, which provides the mechanism linking the air-service shock to tourism labor-market outcomes.

The passenger estimates also show limited route substitution during the nine-month post-bankruptcy window. If other carriers had quickly absorbed WOW Air’s routes, fare segment, and seasonal capacity, the observed series should have moved back toward the counterfactual. Instead, the gap between expected and observed departures persisted across the post-shock window. Figure 2 reports the monthly percentage deviations from the counterfactual. Monthly departures generally remained about 15–25% below expected levels, with no clear return to the counterfactual path before the end of 2019. The persistent gap is consistent with the low-cost carrier literature. Low-fare carriers often expand the market by serving price-sensitive travelers, experimenting with routes, and filling segments that full-service carriers may not immediately replace. In that sense, WOW Air did not simply add interchangeable seats. It appears to have supported a travel segment that was harder to substitute after exit.

3.2 Labor-Market Effects

Table 3 reports whether the passenger shock reached employment, unemployment, and wages. Each row compares an observed Icelandic labor-market or wage outcome with its Bayesian structural time-series counterfactual for April–December 2019. The table is organized in three panels: national labor indicators, employment by sector, and taxable wages by sector. The average-monthly-impact column reports the difference in original units: workers for employment and unemployment, percentage points for the unemployment rate, and ISK

Figure 2: Monthly Passenger Departures Relative to the Counterfactual



Note: The solid line shows the estimated monthly percentage difference between observed and counterfactual departures. Shaded areas represent 95% credible intervals. The table embedded in the figure reports actual, predicted, and difference values for each post-shock month.

millions for wages. The year-over-year column reports the raw observed change from 2018 to 2019, which separates model-based counterfactual effects from simple annual comparisons and helps distinguish relative declines from absolute declines. The RMSE column reports pre-period prediction error. The remaining columns follow the structure used in Table 2.

Table 3: Labor-Market and Wage Effects of the WOW Air Bankruptcy

Outcome	Relative Impact (%)	Avg. Monthly Impact	Monthly Avg. Predicted	Monthly Avg. Observed	YoY Actual Change (%)	p-value	RMSE (pre-period)
National labor indicators							
Unemployed Persons	48.62***	2,613	5,654	8,267	32.62	0.0006	1,263
Unemployment Rate	49.70***	1.30	2.76	4.06	30.82	0.0014	0.66
Employment by sector							
Tourism Employees	-8.11***	-6,010	73,995	67,985	-3.93	0.0002	739
Tourism Employees, Net of Air Transport	-6.59***	-4,498	68,072	63,574	-2.49	0.0010	765
Transportation Employees	-16.51***	-2,653	16,034	13,382	-12.03	0.0002	200
Transportation Employees, Net of Air Transport	-11.94***	-1,217	10,187	8,971	-6.99	0.0002	162
Accommodation Employees	-10.53***	-2,052	19,345	17,293	-4.09	0.0010	567
Air Transport Employees	-25.85***	-1,542	5,953	4,411	-20.76	0.0002	116
Food & Beverage Employees	-8.41***	-939	11,084	10,144	-6.55	0.0002	167
Other Tourism Employees	0.73	169	32,143	32,311	23.09	0.4539	36
Taxable wages by sector, ISK millions							
Taxable Wages, Full Economy	-2.45	-2,916	117,902	114,986	3.20	0.1175	1,233
Tourism Wages	-6.27***	-2,065	32,710	30,644	0.03	0.0042	390
Transportation Wages	-15.64***	-1,564	9,960	8,397	-8.48	0.0002	159
Accommodation Wages	-6.99***	-448	6,319	5,871	1.57	0.0075	131

Note: Average monthly impact is measured in workers for employment and unemployment, percentage points for the unemployment rate, and ISK millions for wages. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The largest employment effects appear in sectors closest to the access shock shown in Figure 1. Air-transport employment predicts an average of 1,542 fewer employees than the counterfactual, higher than the approximately 1,000 employees that would be immediately laid off as a result of WOW Air’s closure.⁹ Tourism employment net of air transport was 6.6% below expected levels, or about 4,498 fewer workers per month, and transportation employment net of air transport was 11.9% below expected levels, or about 1,217 fewer workers per month. The bankruptcy therefore reverberated across the tourism industry more broadly, abruptly halting years of sectoral growth. Transportation services, accommodation, food and beverage, tour services, and other visitor-facing positions depend on the same passenger flows. Appendix Table S7 reports supporting indicators showing reduced rental-car and traffic activity after the bankruptcy. The results do not imply that every sector moved equally; they show that the access shock extended beyond the airline itself.

National unemployment increased as well, both relative to the counterfactual and year over year. The number of unemployed persons was 48.5% higher than expected, or about 2,613 additional unemployed persons per month. The unemployment rate was also nearly 50% above the counterfactual. The percentage effect is large partly because expected unemployment was low, but the level effect is meaningful in a national labor force of roughly 200,000 workers. Figure 4 shows the monthly labor-market deviations from the counterfactual. Although the monthly estimates are noisier than the passenger estimates, the increase in unemployment is visible across the post-bankruptcy period. The tourism employment results also show why year-over-year change and counterfactual change should be read together: tourism employment was expected to continue growing, but instead fell. When growth stops suddenly, the labor effect appears as both realized losses and jobs that would otherwise have been created.

Unemployment did not increase one-for-one with tourism employment shortfalls. Employment adjustment likely occurred through layoffs, reduced hiring, movement into other

⁹See CNBC, “Icelandic airline WOW Air collapses and cancels all flights,” <https://www.cnbc.com/2019/03/28/icelandic-airline-wow-air-collapses-and-cancels-all-flights.html>.

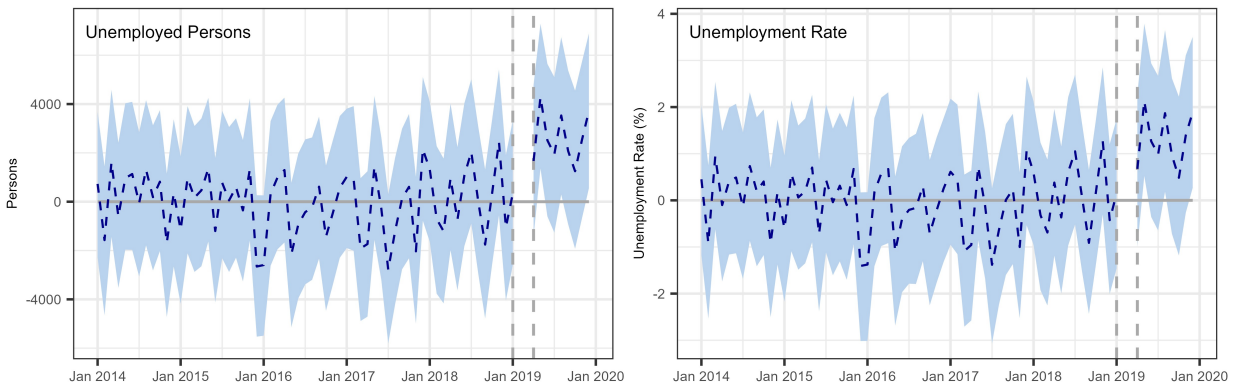
sectors, reduced hours, and exits from measured employment. Wage totals provide a complementary measure because they can respond when hours fall before head counts change. The strongest wage effect appears in transportation, the sector closest to the airline collapse, while the tourism-wage effect shows that the consequences were not limited to head counts. Transportation taxable wages were 15.6% below the counterfactual, equal to an average monthly shortfall of 1,564 million ISK. Over the nine-month post-bankruptcy period, that implies roughly 14.1 billion ISK in missing taxable wages, or about \$114 million in 2019 USD.¹⁰ Tourism taxable wages were 6.2% below the counterfactual, equal to an average monthly shortfall of 2,065 million ISK. Over the same period, that implies roughly 18.6 billion ISK in missing taxable wages, or about \$151 million in 2019 USD. Economy-wide taxable wages were also below expected levels, and the point estimate is larger in absolute terms, but the estimate is noisy and not statistically significant. The table therefore does not show an economy-wide wage collapse. It shows large sectoral wage losses in the parts of the economy closest to the passenger shock.

These wage and unemployment results also give the labor-market shock a public-finance interpretation. The missing taxable wages are not only private income losses for workers and firms. They also represent income that would have generated tax revenue, consumption, and economic activity. At the same time, registered unemployment increased by about 2,613 persons per month relative to the counterfactual, meaning the public sector may have moved from collecting revenue on tourism-related earnings to supporting more workers through unemployment systems. This is not a failure of social insurance; it is the purpose of those systems. But it changes the fiscal position of the shock. An airline bankruptcy can move a destination from net inflows connected to wages and visitor activity toward net outflows connected to labor-market support. Figure 3 illustrates the increase in unemployment during the post-bankruptcy period. The estimates may also understate the full adjustment problem

¹⁰Dollar conversions use an approximate 2019 exchange rate of 0.00810 USD per ISK. The conversion is intended only to make the magnitude more interpretable. The empirical estimates are reported in Icelandic krona.

because the analysis cannot observe whether these wage and employment effects would have persisted beyond 2019 in the absence of the global disruption caused by COVID-19.

Figure 3: *National Unemployment Effects of the WOW Air Bankruptcy*



Note: Panels report observed unemployment outcomes relative to Bayesian structural time-series counterfactuals for April–December 2019. Unemployed persons are measured in persons; the unemployment rate is measured in percentage points.

3.3 Robustness Overview

The appendix reports three sets of validity checks, focusing on the primary passenger outcome because it is the direct measure of the access shock. Placebo tests guard against false positives by applying the same treatment timing to outcomes or countries that should not have been affected. The bankruptcy timing does not produce a statistically significant effect for Icelandic fish landings or for passenger outcomes in Greenland, Denmark, or Norway. None of these should be measurably affected by WOW Air’s bankruptcy. A 2018 pre-period placebo applies the same treatment structure to Iceland one year earlier and produces a smaller, weaker negative effect, suggesting that tourism was cooling before the bankruptcy. Because the preferred 2019 model is trained through January 2019, it incorporates the pre-event slowdown. This does not prove that the entire effect is attributable only to WOW Air, but it reduces the concern that the estimate is simply extrapolating a pre-2019 boom indefinitely.

Sensitivity checks vary the control-country set and covariate grouping. The passenger decline remains large and statistically significant when individual control countries are omitted and when different pools of covariates are used. These checks are important because the comparison set is small and the available indicators differ across countries. The main conclusion does not depend on a single comparison economy or one narrow predictor group.

The appendix also compares the preferred model with simpler alternatives. A model without time-series components produces a larger passenger-loss estimate but worse pre-shock fit. This suggests that the preferred model is more conservative and better aligned with a highly seasonal tourism series. A two-way fixed-effects comparison is less stable: changing the pre-period can switch estimates from negative to positive and can also produce effects for control countries. These checks support the Bayesian structural time-series approach in a rapidly growing destination where parallel trends are unlikely.

3.4 Pre-Shock Exposure

Iceland’s pre-shock specialization helps explain why the labor-market response was so large. Table 4 reports annual averages for 2010, 2018, and 2019. Each row reports a different measure of tourism exposure, including foreign visitors per worker and employment shares in tourism-facing sectors. Monthly foreign visitors per worker rose from 0.24 in 2010 to 0.99 in 2018. Visitors per worker is a simple ratio, but it shows how quickly visitor volume grew relative to the domestic labor force. Tourism employees increased from 7.9% of all employees in 2010 to 14.8% in 2018. Air transport nearly doubled as a share of total employment, and tour services more than tripled. A sudden decline in visitors therefore changes the operating environment for firms that staffed for continued growth. Figure 4 complements the annual exposure table by showing the monthly post-bankruptcy deviations for selected labor outcomes.

Table 4: Tourism Employment Exposure Before and After the WOW Air Bankruptcy

Indicator	2010	2018	2019	% Change, 2010-2018	% Change, 2018-2019
Tourists per Worker					
Monthly Foreign Visitors Per Worker	0.24	0.99	0.85	314.61%	-14.10%
Employment Breakdown By Tourism Sector					
Tourism Employees as % of All Employees	7.88%	14.84%	13.96%	88.41%	-5.97%
Transportation Employees as % of All Employees	5.18%	7.56%	6.85%	45.81%	-9.40%
Air Transport Employees as % of All Employees	1.46%	2.72%	2.28%	86.67%	-16.17%
Accommodation Employees as % of All Employees	5.42%	8.98%	8.66%	65.88%	-3.54%
Food and Bev Employees as % of All Employees	3.91%	5.48%	5.16%	40.41%	-5.84%
Tour Services Employees as % of All Employees	0.60%	1.98%	1.97%	232.45%	-0.49%
Other Tourism Employees as % of All Employees	0.63%	1.45%	1.35%	130.35%	-6.64%

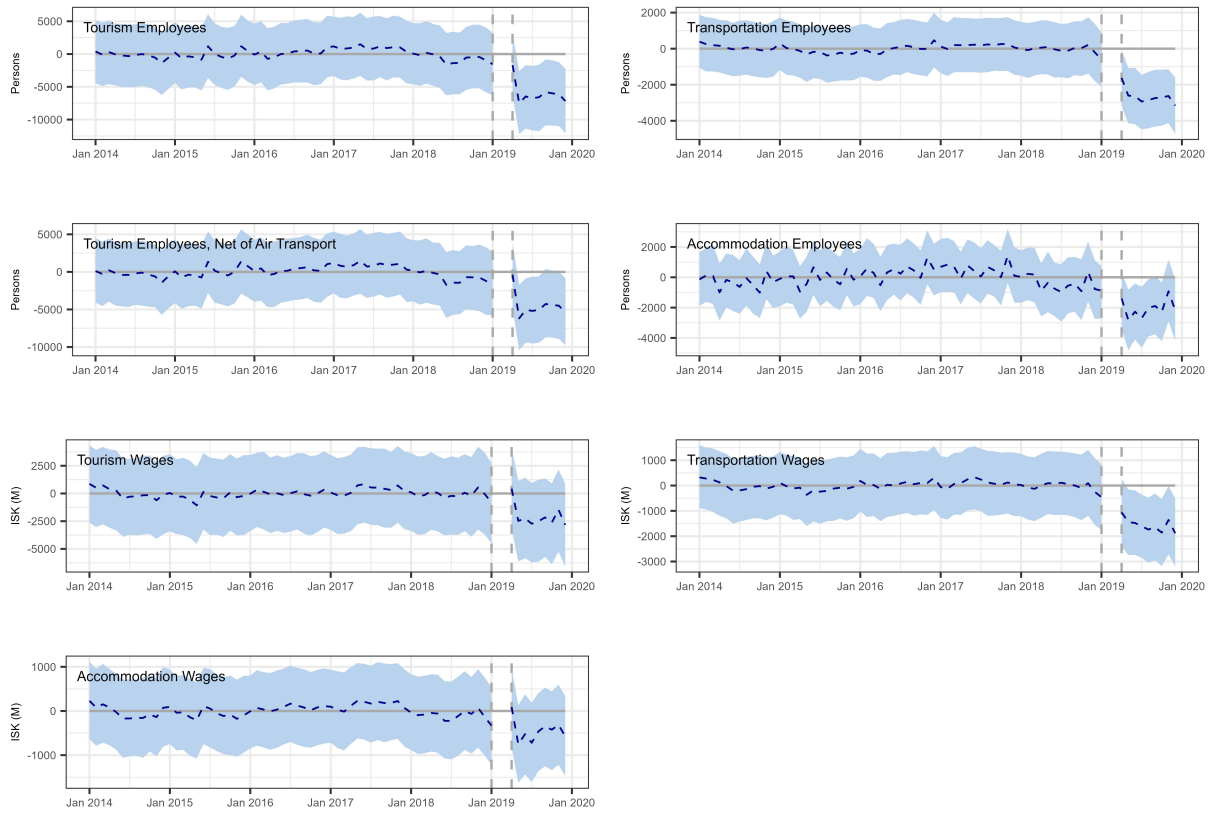
Note: Values are annual averages calculated from monthly data. The first three year columns show the level of each exposure measure. The final two columns show relative changes before the bankruptcy period (2010–2018) and across the first post-bankruptcy year (2018–2019).

Exposure declined immediately after the bankruptcy. Tourism employment as a share of all employment fell from 14.8% in 2018 to 14.0% in 2019, and air-transport employment fell from 2.7% to 2.3%. Iceland entered 2019 with a labor market increasingly specialized in tourism-facing work, and the bankruptcy reduced the passenger flows supporting that specialization.

Despite Iceland’s growth and specialization, the case may still generalize to destinations with concentrated access and tourism-facing labor exposure. Iceland is a small island economy, but it is still more diversified than many island tourism destinations. Several Caribbean and small-island economies have even larger tourism shares of GDP, employment, or export earnings, meaning that the same type of air-access shock could have larger aggregate consequences in more specialized settings. In 2019, travel and tourism accounted for more than half of total employment in each of the ten most tourism-dependent destinations, ranging from 51.3% in Anguilla to 90.7% in Antigua and Barbuda.¹¹ The Iceland case is therefore not an extreme upper bound. It is a case study showing how access vulnerability can become labor-market vulnerability even in a relatively high-income and institutionally capable island economy.

¹¹See World Economic Forum, “These are the 10 places that rely most on tourism for jobs,” <https://www.weforum.org/stories/2020/08/destinations-rely-most-on-tourism-travel/>. The figure is attributed there to WTTC Economic Impact Research; see World Travel & Tourism Council, “Travel & Tourism Economic Impact Research,” <https://wttc.org/Research/Economic-Impact>.

Figure 4: Monthly Differences Between Observed and Counterfactual Labor Levels



Note: Lines report monthly differences between observed labor outcomes and the Bayesian structural time-series counterfactual. Shaded areas indicate posterior uncertainty.

4 Discussion

WOW Air was a low-cost carrier whose growth coincided with, and likely helped enable, Iceland's tourism boom. Its collapse removed low-fare access and suddenly halted that growth. Reduction in foreign visitors produced measurable short-run losses in tourism-facing labor markets. International passenger departures were about 20.6% below the counterfactual from April through December 2019, with foreign travelers accounting for most of the shortfall. Tourism employment was more than 8% below its expected path, and tourism employment net of air transport was still 6.6% lower than expected. Registered unemployment also rose sharply relative to the counterfactual. These results show that tourism exposure can quickly become an economic vulnerability when the transport systems supporting visitor flows are concentrated, fragile, or difficult to replace.

The distinction between exposure and vulnerability matters for policymakers. Exposure may increase as a region's share of tourism exports increase. But the associated risk may come from oversight of the wider systems supporting tourism. A destination can remain attractive while still experiencing a sharp decline in realized visitor traffic if the transportation that connects visitors to the destination is disrupted. In this sense, the bankruptcy was different from many tourism shocks studied in the literature. Pandemics, terrorism, natural disasters, and political crises often affect whether visitors want to travel. WOW Air's collapse primarily affected whether visitors could reach Iceland at the same scale, price, and route structure. The destination did not suddenly lose its appeal. Rather, a major part of the supply chain supporting recent tourism growth disappeared. The evidence therefore supports a narrow but important claim, that visitor traffic is not only demand-dependent. It is also dependent on accessibility.

The resulting labor-market outcomes also show that shock is not self-contained. The most immediate and direct losses are understandably connected to the failed airline and the air-transport sector. However, effects were not limited to the airline itself. Transportation, broader tourism employment, sectoral wages, and registered unemployment also increased

substantially relative to the counterfactual. For a destination, the relevant issue is not only how many airline employees lose their jobs. It is also whether hotels, tour operators, restaurants, ground transportation firms, and other visitor-facing businesses face fewer customers, reduced hours, slower hiring, and lower wage payments. The results indicate this is a likely outcome.

In a more optimistic note, the findings also show why the WOW Air case should not be viewed solely as a story of fragility. WOW Air appears to have contributed meaningfully to Iceland's pre-2019 tourism growth by expanding low-fare access, connecting markets, and helping sustain a larger visitor economy. That growth created employment and supported a more specialized tourism sector. When the carrier disappeared, other airlines did not immediately replace the missing capacity, and the labor market moved well below its expected path. The policy lesson is therefore not that low-cost carrier growth is undesirable. The more useful conclusion is that carrier-led growth should be paired with planning for what happens if that capacity disappears. More broadly, destinations need to know when visitor flows depend on a small number of private carriers, routes, or fare segments.

The policy implication is therefore not airline rescue, but exposure planning. The results show that the negative tourism shock had both private labor-market consequences and a public-finance dimension. Taxable wages fell in affected sectors while registered unemployment increased, a combination that creates a political and financial opening for soft-landing policies aimed at affected workers, firms, and sectors, while still stopping well short of an airline bailout. Destination agencies should understand which systems of the visitor economy depend on concentrated or fragile access. That includes the share of seats provided by a major carrier, the origin markets served by few airlines, the role of low-fare capacity, the seasonality of routes, and the sectors most tied to the affected visitor segments. If tourism grows faster than other industries, it is increasing reliance on these systems, and it is important to plan accordingly.

Several limitations should be noted. The analysis estimates short-run effects from April

through December 2019 and does not measure longer-run recovery because COVID-19 disrupted the subsequent period. It cannot fully distinguish layoffs from foregone hiring or movement between sectors. The comparison economies are imperfect controls, but that limitation is also why the Bayesian structural time-series design is useful, by using imperfect but informative indicators without requiring identical outcome definitions across countries. The estimates should be interpreted as deviations from a synthetic counterfactual conditional on pre-bankruptcy information, not as a claim that all 2019 tourism cooling was caused by WOW Air. These limitations narrow the claim, but they do not eliminate the central finding that the loss of a single airline can reduce destination accessibility and produce measurable short-run impacts extending beyond the airline itself.

The implications of this empirical design should not be viewed solely as a case study. By comparing observed outcomes with a synthetic no-bankruptcy path, the analysis shows that transport disruptions result in both visitor-flow shocks and, consequently, labor-market shocks. For small, island, and peripheral destinations, tourism development strategies should encourage resilience addressing connectivity to limit their vulnerability.

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Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this manuscript, the author used OpenAI's ChatGPT for editorial assistance, including language revision, organization, and formatting suggestions. The author reviewed and completed the analysis, interpretation, and final manuscript content and takes full responsibility for the work.

References

- Abadie, Alberto, Alexis Diamond, and Jens Hainmueller.** 2010. “Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California’s Tobacco Control Program.” *Journal of the American Statistical Association*, 105(490): 493–505.
- Andrianambinina, F. Ollier D., Derek Schuurman, Mamy A. Rakotoarijaona, Chantal N. Razanajovy, Honorath M. Ramparany, Serge C. Rafanoharana, H. Andry Rasamuel, Kevin D. Faragher, Patrick O. Waeber, and Lucienne Wilme.** 2023. “Boost the resilience of protected areas to shocks by reducing their dependency on tourism.” *PLOS ONE*, 18(4): e0278591.
- Bachwich, Alexander R., and Michael D. Wittman.** 2017. “The emergence and effects of the ultra-low cost carrier business model in the U.S. airline industry.” *Journal of Air Transport Management*, 62: 155–164.
- Barrett, Sean D.** 2004. “How do the demands for airport services differ between full-service carriers and low-cost carriers?” *Journal of Air Transport Management*, 10(1): 33–39.
- Brodersen, Kay H., Fabian Gallusser, Jim Koehler, Nicolas Remy, and Steven L. Scott.** 2015. “Inferring causal impact using Bayesian structural time-series models.” *The Annals of Applied Statistics*, 9: 247–274.
- Canh, Nguyen Phuc, and Su Dinh Thanh.** 2020. “Domestic tourism spending and economic vulnerability.” *Annals of Tourism Research*, 85: 103063.
- Chung, Jin Young, and Taehee Whang.** 2011. “The impact of low cost carriers on Korean Island tourism.” *Journal of Transport Geography*, 19(6): 1335–1340.
- Duro, Juan Antonio, Alejandro Perez-Laborda, Judith Turrion-Prats, and Melchor Fernández.** 2021. “Covid-19 and tourism vulnerability.” *Tourism Management Perspectives*, 38: 100819.

- Faber, Benjamin, and Cecile Gaubert.** 2019. "Tourism and Economic Development: Evidence from Mexico's Coastline." *American Economic Review*, 109: 2245–2293.
- Graham, Anne.** 2013. "Understanding the low cost carrier and airport relationship: A critical analysis of the salient issues." *Tourism Management*, 36: 66–76.
- Gwenhure, Yvonne, and Nicholas M Odhiambo.** 2017. "Tourism and economic growth: A review of international literature." *Tourism (13327461)*, 65: 33–44.
- Hong Tsui, Kan Wai.** 2017. "Does a low-cost carrier lead the domestic tourism demand and growth of New Zealand?" *Tourism Management*, 60: 390–403.
- Kocak, Emrah, Umit Bulut, and Selim Bakir.** 2025. "Do shocks to tourism receipts have a transitory or persistent nature? A comparison of traditional and artificial intelligence-based analytical procedures." *Tourism Economics*, 32(4): 770–781.
- Kumar, Ronald Ravinesh, and Peter Josef Stauvermann.** 2021. "Tourism and economic growth in the Pacific region: evidence from five small island economies." *Journal of the Asia Pacific Economy*, 28: 1–28.
- McCarroll, Meghan J., G. Thomas LaVanchy, and Michael W. Kerwin.** 2024. "Tourism resilience to drought and climate shocks: The role of tourist water literacy in hotel management." *Annals of Tourism Research Empirical Insights*, 5: 100147.
- Mirkova, Ivelina, and Noemi Padrón-Fumero.** 2025. "From vulnerability to resilience: Empowering stakeholder-driven just transitions in island tourism economies." *Energy Research & Social Science*, 121: 103966.
- Morakabati, Yeganeh.** 2019. "A question of confidence. Is tourism as vulnerable to civil unrest as we think? A comparative analysis of the impact of Arab Spring on total reserves and tourism receipts." *International Journal of Tourism Research*, 22: 252–265.

- Nguyen, Canh Phuc, and Thanh Dinh Su.** 2021. “The Vulnerability Effect of International Tourism on a Destination’s Economy.” *Tourism Planning & Development*, 1–21.
- Payne, James E., and Saban Nazlioglu.** 2023. “The permanent or transitory nature of shocks to tourism expenditures and receipts: Evidence from new panel stationarity tests with breaks and factors.” *Tourism Economics*, 29(6): 1518–1532.
- Perles-Ribes, José Francisco, Ana Belén Ramón-Rodríguez, María Jesús Such-Devesa, and Patricia Aranda-Cuéllar.** 2023. “The Immediate Impact of Covid19 on Tourism Employment in Spain: Debunking the Myth of Job Precariousness?” *Tourism Planning & Development*, 20(1): 1–11.
- Pulina, Manuela, and Isabel Cortés-Jiménez.** 2010. “Have Low-Cost Carriers Influenced Tourism Demand and Supply? The Case of Alghero, Italy.” *Tourism Analysis*, 15: 617–635.
- Roudi, Samira, Huseyin Arasli, and Seyi Saint Akadiri.** 2018. “New insights into an old issue – examining the influence of tourism on economic growth: evidence from selected small island developing states.” *Current Issues in Tourism*, 22: 1280–1300.
- Saleh, Ali Salman, Reetu Verma, and Ranjith Ihalanayake.** 2011. “Do External Shocks Have a Permanent or a Transitory Effect on Thailand’s Tourism Industry?” *Tourism Analysis*, 16: 483–491.
- Santos, Anabela, and Michele Cincera.** 2018. “Tourism demand, low cost carriers and European institutions: The case of Brussels.” *Journal of Transport Geography*, 73: 163–171.
- Seetanah, B.** 2011. “Assessing the dynamic economic impact of tourism for island economies.” *Annals of Tourism Research*, 38: 291–308.

- Sun, Xiaoqian, Changhong Zheng, Kun Wang, and Sebastian Wandelt.** 2025. “The art of airline cessation: A comprehensive analysis based on airline ceases reported between the years 2010 and 2023.” *Journal of Air Transport Management*, 124: 102748.
- Sun, Ya-Yen, Lintje Sie, Futu Faturay, Ilmiawan Auwalin, and Jie Wang.** 2021. “Who are vulnerable in a tourism crisis? A tourism employment vulnerability analysis for the COVID-19 management.” *Journal of Hospitality and Tourism Management*, 49: 304–308.
- Sun, Ya-Yen, Mengyu Li, Manfred Lenzen, Arunima Malik, and Francesco Pomponi.** 2022. “Tourism, job vulnerability and income inequality during the COVID-19 pandemic: A global perspective.” *Annals of Tourism Research Empirical Insights*, 3(1): 100046.
- Wang, Yong, Linna Han, and Xuejiao Ma.** 2022. “International tourism and economic vulnerability.” *Annals of Tourism Research*, 94: 103388.
- Yucel, Ali Gokhan.** 2020. “Are shocks to tourist arrivals permanent or transitory? A comprehensive analysis on the top 20 most-visited countries.” *Current Issues in Tourism*, 24: 1–18.
- Zhang, Hongru, and Yang Yang.** 2025. “Tourism shocks and gender unemployment dynamics: Unraveling the gender gap in Macau’s labor market.” *Tourism Economics*.

Appendix

A Data Availability and Model Calibration

This section provides additional detail on data availability, covariate selection, and pre-period calibration. The method used in the main analysis was selected to address the need for monthly data and the uneven availability of comparable indicators across small North Atlantic and Nordic economies. The tables and figures below show why a flexible Bayesian structural time-series design is useful in this setting and how the preferred model is calibrated.

Table S1 reports the availability of monthly indicators by topic area and comparison economy. The first column lists the measure. The remaining columns show whether the measure is available for Iceland, the Faroe Islands, Greenland, Denmark, and Norway, and whether it is available as an aggregate or with additional detail. The table illustrates why conventional synthetic control methods are difficult in this setting. In the standard synthetic-control framework, the treated outcome is approximated by a weighted combination of the same outcome observed in control units (Abadie, Diamond and Hainmueller, 2010). That structure would require, for example, air-transport employment in Iceland to be predicted using comparable air-transport employment series in other countries. Table S1 shows why that requirement is restrictive. Iceland reports detailed sectoral tourism and labor outcomes, while some comparison economies report only aggregate indicators. The Bayesian structural time-series model addresses this limitation by allowing all available indicators to contribute to prediction when they improve pre-shock fit.

Figure S1 shows the share of potential covariates in the compiled dataset that are observed without missing values over time. Data coverage improves sharply by 2014. As a result, the preferred model training period begins in 2014 because this start date balances the need for enough pre-shock time-series information with the need for a rich set of monthly comparison indicators.

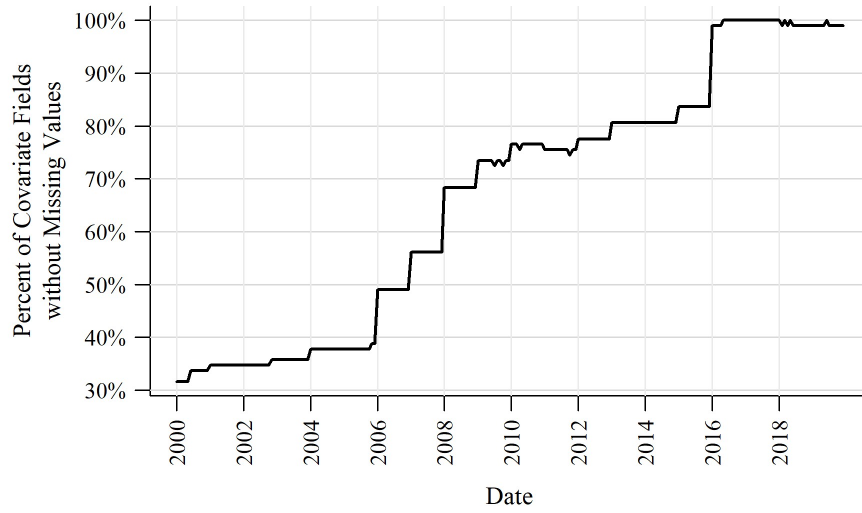
Figure S2 reports cumulative passenger effects across alternative pre-period start dates.

Table S1: Varying Data & Information Availability

Measure	Icelandic	Faroe Islands	Greenland	Denmark	Norway
Tourism & Transportation					
Airport Passengers	Arrivals, Departures by Citizenship	Arrivals	International, Total	Passengers	Passengers
Cruise Passengers	Total	-	Total	-	-
Hotel Occupancy Rate	By Bed, Room, by Type	-	-	Total	Total
Hotel Check-ins	Total, Foreign, Domestic, by Type	Total (Room Nights)	Total (Guests)	-	Total
Overnight Stays	Total, Foreign, Domestic, by Type	Total	Total	Total, Foreign	Guest Nights (Total, Foreign)
Hotel Rooms Available	-	-	-	Total	-
Rental Car Registrations	Total	-	-	-	-
Labor					
Unemployment Rate	Total, By Age	Total, By Age	Total	Total, By Age Group	Total, By Age
Unemployed Persons	Total, By Age	By Age	Total, Job Seekers Total, Under 25	Total, By Age Group	Total, By Age
Employed Persons	Total, By Age, By Sector	Total, By General Industry	Total, By Sector	Total, By Sector	Total Industry, By Sector, By Age
Wages	Total By Sector	Total, By Sector	-	-	Total Industry, By Sector
Hours Worked	Total, By Age	-	-	-	-
Insolvencies	Total, By Sector	-	-	Total, By Sector	Total, By Sector
Business Openings	-	-	-	Total, By Sector	-
National Accounting					
CPI	Total	-	-	Total	Total
Exports	Total	Total	Total	Total	Total
Imports	Total	Total	Total	Total	Total
Unemployment Rate	Total	Total	Total	Total	Total
Employment Counts	Total	Total	Industry Total	Industry Total	Industry Total
Insolvencies	Total	-	-	Total, By Sector	Total, By Sector
Business Creation	-	-	-	Total, By Sector	-
Other					
Local Traffic Count	Total, by Region	-	-	-	-
Fuel Use	Total, Paid by Foreign, Domestic	-	-	Total	Total
Fish Landings	Weight	By Weight / Currency	By Weight / Currency	-	Weight

Note: Reported measures by country showing variation in availability and level of detail across topic areas.

Figure S1: Monthly Covariate Series with Complete Data Coverage

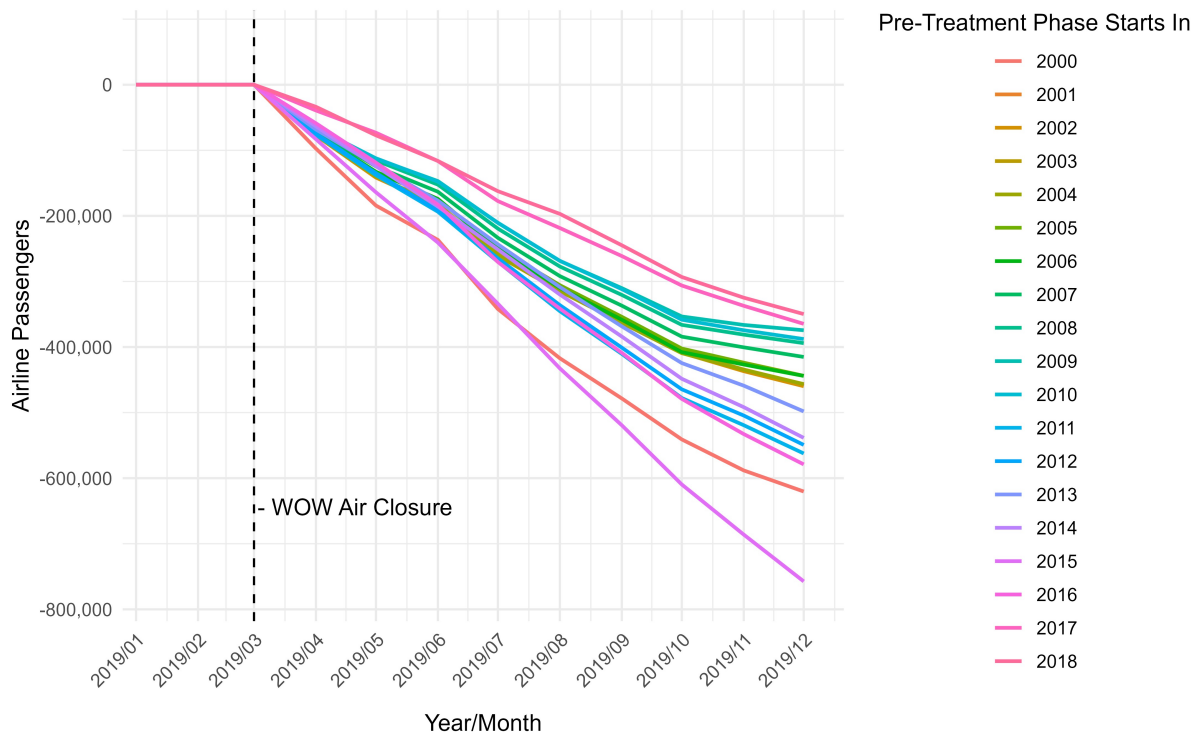


Note: The figure reports the percentage of candidate covariate fields with non-missing monthly values by date. Higher coverage allows richer counterfactual prediction.

The calibration exercise supports using 2014 as the main training start. Beginning much earlier provides more pre-period observations but substantially reduces available covariate data and reaches further back into a different period of Icelandic tourism development. Beginning later provides richer covariate coverage but gives the model fewer pre-shock observations. The 2014 start balances those concerns and produces an estimate close to the center of the calibration distribution. The estimated passenger decline remains negative across the calibration range, and the preferred 2014 specification yields an estimated cumulative decline of about 538,000 departing passengers, very similar to the main estimate. All other hyperparameters follow the defaults recommended by Brodersen et al. (2015) unless otherwise noted.

Figure S3 reports posterior inclusion probabilities for the primary passenger model. Each bar indicates how often a covariate is selected across posterior draws. Higher posterior inclusion probabilities identify indicators that are more consistently useful in predicting Iceland's pre-shock passenger trajectory. The figure also reports coefficient information, so it should be read as both a selection diagnostic and a sign/magnitude check. A high

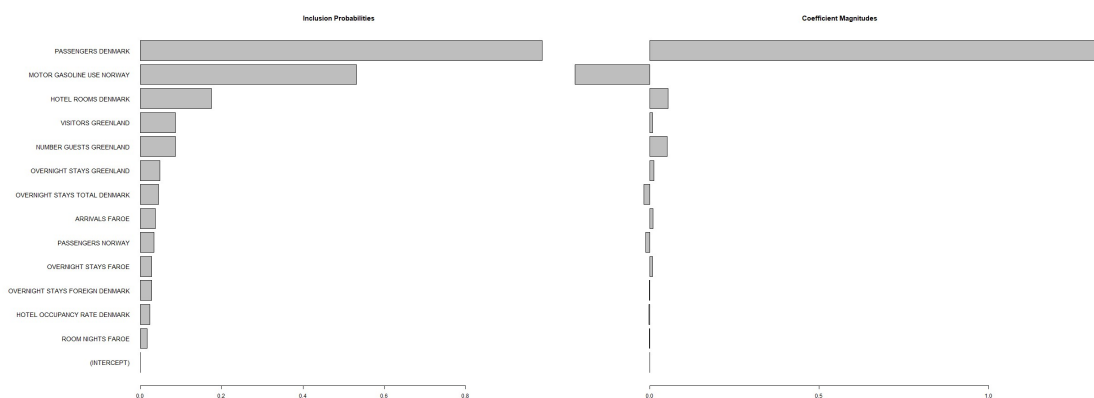
Figure S2: Cumulative Passenger Effects Across Alternative Training Windows



Note: Each point reports the cumulative April–December 2019 passenger effect estimated using a different pre-period start date. The median cumulative change across tested windows is approximately -458,000 passengers, the mean is approximately -487,000 passengers, and the 2014 specification is approximately -538,000 passengers.

inclusion probability with a small coefficient can indicate a stable but modest predictor; a lower inclusion probability with a larger coefficient can indicate a predictor that is useful in some posterior draws but not consistently selected. This selection process helps address overfitting because weak predictors are shrunk toward zero rather than being forced into the counterfactual.

Figure S3: Posterior Inclusion Probabilities for the Passenger Model



Note: Posterior inclusion probabilities indicate how often each candidate predictor is selected by the spike-and-slab prior during estimation. Predictors with higher values contribute more consistently to the pre-shock counterfactual fit.

B Placebo Checks

Placebo checks test whether the model detects effects where none should be expected. Table S2 reports three categories of placebo tests. The first panel applies the bankruptcy timing to an unrelated Icelandic outcome, fish landings. The second applies the treatment timing to other countries. The third applies the treatment to earlier Icelandic periods before the bankruptcy.

The placebo results are reassuring because the model does not produce comparable effects in places or outcomes where the bankruptcy should not have had a direct impact. The fish-landings placebo is statistically insignificant, indicating that the model does not generate a false effect for an Icelandic sector unrelated to international air access. Placebos applied to Greenland, Denmark, and Norway are also statistically insignificant. The Faroe Islands estimate is larger in magnitude but statistically imprecise, consistent with volatility in a very small passenger market. The 2018 placebo is negative and weakly significant, which may reflect a cooling of Icelandic tourism before the bankruptcy. That pre-period placebo is important because it cautions against attributing every element of the 2019 slowdown to WOW Air. However, it is smaller and less precise than the 2019 bankruptcy effect, and the preferred model is trained through January 2019 so that information from the pre-bankruptcy cooling period contributes to the counterfactual.

C Robustness Checks

The robustness checks vary the control-country set, covariate grouping, and time-series structure of the model. The purpose is to evaluate whether the passenger result depends on a particular modeling choice.

Table S2: Placebo Counterfactual Tests

Indicator	Relative Impact (%)	9-Month Cumulative Impact	95% CI Upper	95% CI Lower	p-value	RMSE (priors)
Total Passengers	-20.63***	-533,643	-251,386	-942,546	0.0002	13,530
Icelandic Passengers	-16.70***	-100,771	-23,111	-180,957	0.0010	2,905
Foreign Passengers	-22.75***	-461,168	-211,729	-831,090	0.0002	11,895
Fish Landings	-2.88	-32,226,026	135,201,145	-187,612,889	0.3374	23,174,309
Greenland	-0.14	-1,888	19,389	-26,211	0.4369	229
Faroe Islands	8.97	23,269	68,348	-33,982	0.1401	NA
Denmark	-3.20	-449	432	-1,285	0.1606	41
Norway	1.85	166,696	744,650	-555,206	0.2694	21,952
Impact in January 2018	-7.48*	-198,384	79,105	-454,292	0.0742	11,081
Impact in January 2017	8.32	146,501	486,210	-438,857	0.2444	11,023

Note: The first column lists each placebo or treated series. Relative impact is the estimated post-treatment deviation from the counterfactual as a percentage of the counterfactual. The cumulative-impact column reports the total post-treatment difference in original units. Placebo effects that are small or statistically insignificant support the interpretation that the main Iceland passenger effect is not a general modeling artifact. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.1 Control-Country Selection

Table S3 omits one or more comparison countries from the preferred passenger model. The estimated passenger decline remains large and statistically significant across all specifications. The relative impact ranges from about -20% to -29%, with an average of -23.4%.

Table S3: Passenger Effects Under Alternative Control-Country Sets

Omitted Control Country	Relative Impact (%)	9-Month Cumulative Impact	95% CI Upper	95% CI Lower	p-value	RMSE (priors)
Greenland	-20.80***	-541088	-246711	-945642	4e-04	13870
Faroe Islands	-22.28***	-592873	-269139	-1088169	2e-04	12945
Denmark	-27.98***	-791612	-477276	-1092558	2e-04	19983
Norway	-20.38***	-531946	-214080	-1013803	8e-04	14075
Faroe Islands & Greenland	-21.47***	-561587	-255464	-956970	2e-04	13839
Greenland & Denmark	-28.77***	-820965	-537468	-1125084	2e-04	20322
Denmark & Faroe Islands	-21.15***	-545757	-272972	-817773	2e-04	22257
Denmark & Norway	-26.30***	-727234	-420244	-1024210	2e-04	23640
Norway & Faroe Islands	-20.97***	-552702	-217453	-1013604	2e-04	13780
Mean	-23.34***	-629529	-323423	-1008646	3e-04	17190

Note: Each row reports the main passenger model after omitting the listed control country or countries. The stability of the negative effect indicates that the main result is not driven by a single comparison economy. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.2 Covariate Grouping

Table S4 varies the topic area of indicators used to estimate the counterfactual. The first column lists the covariate group. The estimated passenger decline remains statistically significant across all groups. The preferred tourism-indicator model balances substantive relevance and pre-period fit, but the direction and approximate scale of the effect are robust to broader or narrower predictor sets.

Table S4: Passenger Effects Under Alternative Covariate Groups

Indicator	Relative Impact (%)	9-Month Cumulative Impact	95% CI Upper	95% CI Lower	p-value	RMSE (priors)
All Indicators	-22.30***	-583478	-359199	-801125	2e-04	11422
Passenger Indicators	-17.56***	-435721	-210164	-692980	4e-04	15277
Tourism Indicators	-20.59***	-533643	-246692	-945415	2e-04	13530
Labor Indicators	-22.15***	-578100	-350623	-792190	2e-04	11089
Other Indicators	-23.69***	-630521	-361031	-896049	2e-04	26532
Mean	-21.26***	-552293	-305542	-825552	2e-04	15570

Note: Each row reports the passenger model estimated with a different group of candidate predictors. Relative impact is the estimated April–December 2019 passenger deviation from the counterfactual divided by the counterfactual. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.3 Counterfactual Without Time-Series Components

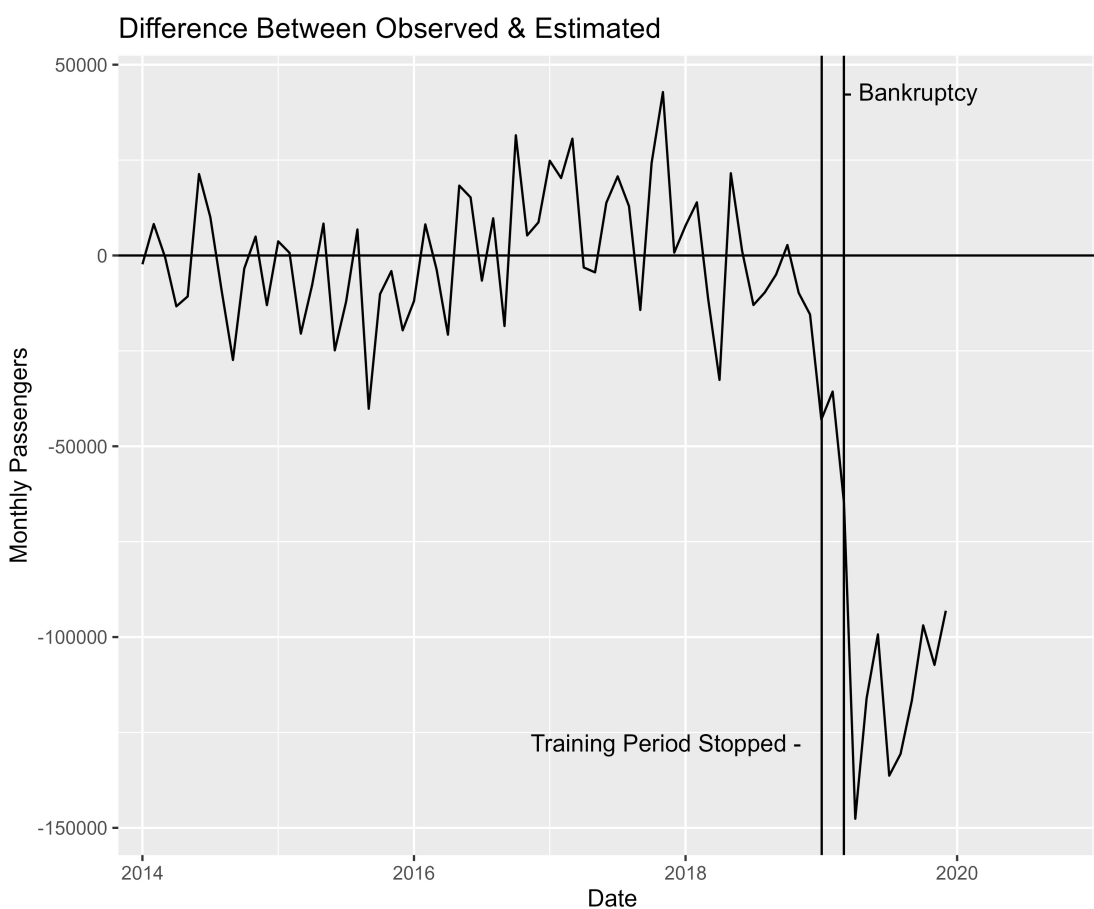
Table S5 compares the preferred Bayesian structural time-series model with a simplified regression that excludes time-series and seasonality components. The simplified model produces a larger estimated effect but a worse pre-period fit. This indicates that the preferred model is more conservative and better suited to a tourism outcome with strong seasonality and a changing trend.

Table S5: Passenger Effects With and Without Time-Series Components

Model Used	Loss Percent	Monthly Impact	Cumulative Impact	P-Value	RMSPE
Simple OLS Regression, Time Series Omitted	-34.01%***	-115,986	-1,043,878	5.2e-05	17,204
BSTS Preferred Model	-20.63%***	-59,294	-533,643	2e-04	13,530

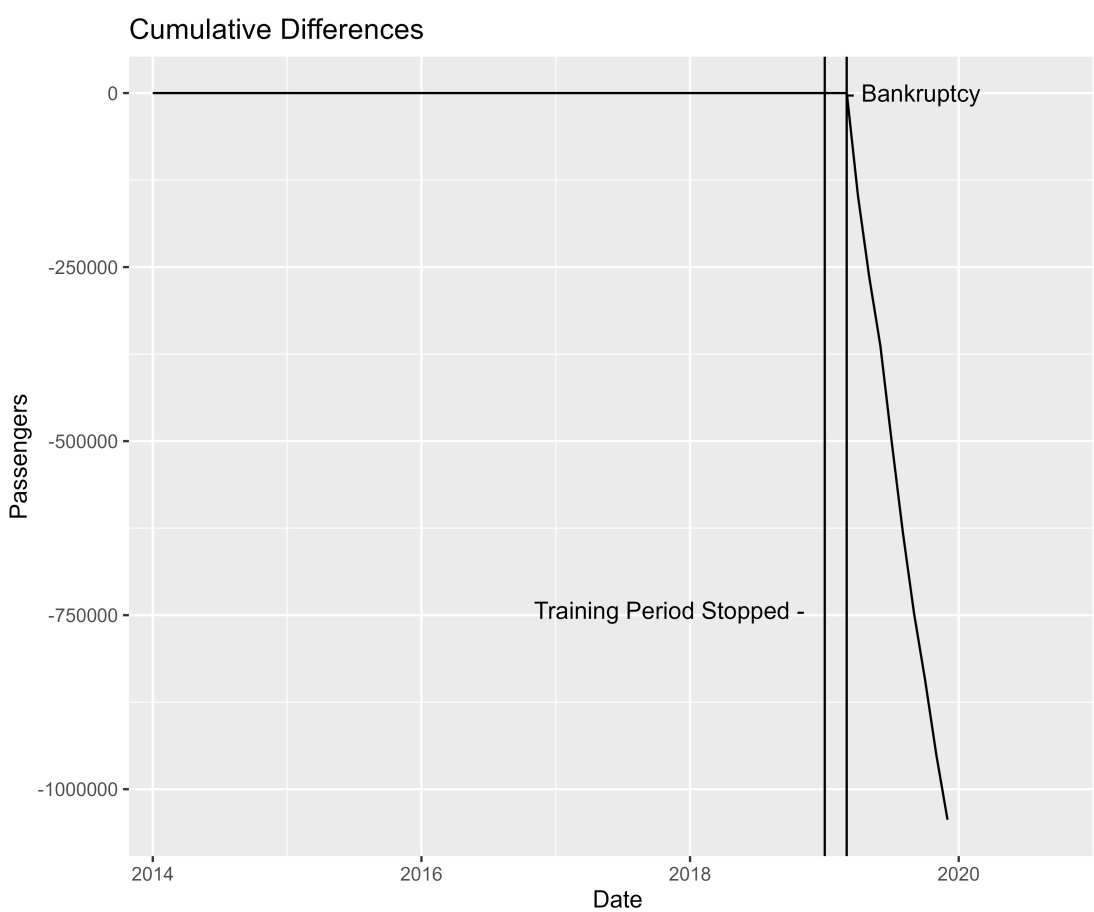
Note: The first row reports a simplified regression using contemporaneous comparison indicators without local trend or seasonal components. The second row reports the preferred Bayesian structural time-series specification. RMSPE is the pre-period prediction error. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure S4: Monthly Passenger Differences Without Time-Series Components



Note: The figure reports monthly differences between observed departures and a simplified counterfactual that excludes time-series components. The simplified model produces larger errors than the preferred specification.

Figure S5: Cumulative Passenger Differences Without Time-Series Components



Note: The figure reports cumulative differences between observed departures and a simplified counterfactual that excludes time-series components.

D Two-Way Fixed-Effects Comparison

As a benchmark, a two-way fixed-effects model is estimated using log arrivals as the outcome:

$$\begin{aligned} \log(Y_{it}) = & \beta_0 + \beta_1 Treat_i + \beta_2 (Post_t \times Treat_i) \\ & + \beta_3 CountryFE_i + \beta_4 YearFE_t + \beta_5 MonthFE_t + \epsilon_{it}. \end{aligned}$$

The parallel-trends assumption is unlikely to hold because Iceland’s tourism sector grew rapidly before the bankruptcy. Table S6 confirms this concern. Estimated treatment effects for Iceland vary sharply with the length of the pre-period and become positive when more years are included. Some control-country placebos also become statistically significant. These patterns support the use of the Bayesian structural time-series model in the main analysis.

Table S6: Two-Way Fixed-Effects Treatment Estimates Across Pre-Period Windows

	1 year prior	2 years prior	3 years prior	4 years prior	5 years prior	6 years prior
Iceland	-0.2177*	-0.1527**	-0.0641	0.0493	0.1494**	0.2396***
Greenland	0.1395	0.03	-0.0128	-0.0438	-0.0629	-0.0898
Faroe Islands	0.2512**	0.1861***	0.2032***	0.1919***	0.1885***	0.1832***
Norway	-0.0706	-0.0245	-0.0551	-0.1054*	-0.158**	-0.1981***
Denmark	-0.1025	-0.0389	-0.0712	-0.0919*	-0.117*	-0.1349**

Note: Each cell reports the estimated interaction between the treated country and the post-bankruptcy period using a different pre-period length. Rows identify the country assigned to treatment. Instability across pre-period windows and significant placebo effects indicate that two-way fixed effects are poorly suited to this setting. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

E Additional Outcome Tables

Table S7 reports additional activity outcomes connected to visitor movement and local tourism demand. The rows use the same counterfactual setup as the main tables and estimate effects for capital-area road traffic, registered rental cars, and overnight stays.

Table S7: Additional Tourism Activity Effects of the WOW Air Bankruptcy

Outcome	Relative Impact (%)	Avg. Monthly Impact	Monthly Avg. Predicted	Monthly Avg. Observed	YoY Actual Change (%)	p-value	RMSE (pre-period)
Capital-Area Road Traffic	-8.48***	-4,073	47,637	43,564	0.99	0.0051	822
Registered Rental Cars	-20.88***	-6,653	31,499	24,846	-3.63	0.0004	969
Overnight Stays	-4.28	-18,325	410,358	392,033	2.34	0.1680	19,523

Note: Rows report counterfactual estimates for Icelandic visitor-adjacent activity measures. Capital-area road traffic measures road traffic in the capital region; registered rental cars measures the rental-car stock; overnight stays measures overnight stays in Iceland. Average monthly impact is measured in the original units of each row. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.